

Architecture Decision Records (ADRs)

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This document captures key architectural decisions made during the AWS Modernization Platform project.

ADR-001: Why Aurora Global Database Instead of RDS MySQL Cross-Region Read Replica?

Status

Accepted

Context

The modernization platform requires a highly available database architecture capable of supporting disaster recovery across AWS Regions. The application serves business-critical workloads and must minimize both data loss and recovery time during a regional outage.

Options evaluated:

- Amazon RDS MySQL Cross-Region Read Replicas
- Amazon Aurora Global Database

Decision

Amazon Aurora Global Database was selected as the preferred database architecture.

RPO Analysis

RDS Cross-Region Replicas rely on asynchronous binlog replication and may experience several minutes of replication lag during failures.

Aurora Global Database performs storage-level replication with typical replication latency below one second, resulting in near-zero data loss.

RTO Analysis

RDS replicas typically require manual promotion and application reconfiguration, resulting in recovery times of 5–30 minutes.

Aurora Global Database supports managed global failover and typically achieves recovery in under one minute.

Consequences

Positive:

- Near-zero data loss
- Faster disaster recovery
- Better global scalability
- Enterprise-grade resilience

Negative:

- Higher cost
- Additional architectural complexity

Conclusion

Aurora Global Database was selected because it provides significantly better disaster recovery characteristics than RDS MySQL Cross-Region Replicas. The improved RPO and RTO justify the additional cost for business-critical workloads.

ADR-002: Why Amazon EKS Instead of Amazon ECS?

Status

Accepted

Context

The modernization platform requires a container orchestration platform capable of supporting microservices, CI/CD, auto scaling, high availability, and future portability.

Options evaluated:

- Amazon ECS
- Amazon EKS

Decision

Amazon EKS was selected as the primary container orchestration platform.

Evaluation

Amazon EKS provides Kubernetes-based orchestration, industry-standard skills, portability across cloud providers, and access to a rich ecosystem including Helm, ArgoCD, Istio, Prometheus, and Grafana.

Amazon ECS offers simpler operations and tighter AWS integration but is more AWS-specific.

Business Justification

EKS aligns better with enterprise adoption trends and provides stronger evidence of Kubernetes and DevOps expertise, making it more valuable for modernization initiatives and professional development.

Consequences

Positive:

- Kubernetes expertise
- Cloud portability
- Enterprise alignment
- Strong DevOps skill demonstration

Negative:

- Higher complexity
- Additional control plane costs

Conclusion

Amazon EKS was selected because the project prioritizes enterprise-grade architecture, portability, and Kubernetes expertise over operational simplicity.